

Compo Electrostatico : Ley de Coulomb

q



$$\vec{E}(\vec{r}) = k \frac{q (\vec{r} - \vec{r}_q)}{|\vec{r} - \vec{r}_q|^3}$$

Nq_i



$$\vec{E}(\vec{r}) = k \sum_{i=1}^N \frac{q_i (\vec{r} - \vec{r}_{q_i})}{|\vec{r} - \vec{r}_{q_i}|^3}$$



Distr.
Continua de
Cargas

$$\vec{E}(\vec{r}) = k \int \frac{dq (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Punto Campo

Punto Fuente

Densidad de Carga

3D

Volumetrica : $[\rho] = C/m^3$

2D

Superficiales : $[\sigma] = C/m^2$

1D

Lineales : $[\lambda] = C/m$

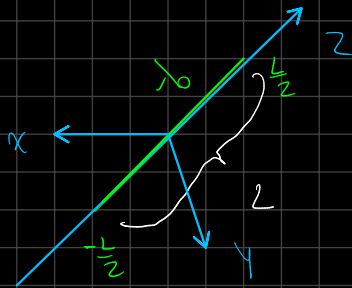
Uniformes

$\rho_0, \sigma_0, \lambda_0$

No Uniformes

$\rho(\vec{r}'), \sigma(\vec{r}'), \lambda(\vec{r}')$

Ejercicio Particular



$$\vec{E}(\vec{r}) = ?$$

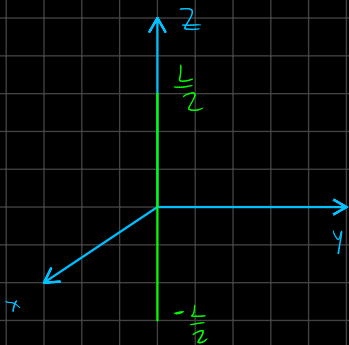
$$\vec{E}(\vec{r}) = k \int \frac{dq' (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Tomamos $dq' = \lambda_0 d\ell' = \lambda_0 dz'$

pto campo. $\vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$

pte Fuente. $\vec{r}' = 0 \hat{x} + 0 \hat{y} + z' \hat{z}$

$$-\frac{L}{2} \leq z' \leq \frac{L}{2}$$



Tomamos ...

$$\vec{E}(\vec{r}) = k \int_{-L/2}^{L/2} \frac{\lambda_0 dz' (x \hat{x} + y \hat{y} + (z - z') \hat{z})}{[x^2 + y^2 + (z - z')^2]^{3/2}} \begin{matrix} \uparrow \rightarrow E_x \\ \rightarrow E_y \\ \downarrow \rightarrow E_z \end{matrix}$$

$$E_x = k \int_{-L/2}^{L/2} \frac{\lambda_0 dz' x \hat{x}}{[x^2 + y^2 + (z - z')^2]^{3/2}} = k \lambda_0 \int_{-L/2}^{L/2} \frac{x dz'}{[x^2 + y^2 + (z - z')^2]^{3/2}}$$

$$E_y = k \lambda_0 \int_{-L/2}^{L/2} \frac{y dz'}{[x^2 + y^2 + (z - z')^2]^{3/2}}$$

$$E_z = k \lambda_0 \int_{-L/2}^{L/2} \frac{(z - z') dz'}{[x^2 + y^2 + (z - z')^2]^{3/2}}$$

Analizando $E_x = k \lambda_0 \times \int_{-L/2}^{L/2} \frac{dz'}{[x^2 + y^2 + (z - z')^2]^{3/2}} = k \lambda_0 \times \int_{z+L/2}^{z-L/2} \frac{-dv}{[x^2 + y^2 + v^2]^{3/2}}$

$v = z - z'$
 $dv = -dz'$

$v_2 = z - L/2 \quad \wedge \quad v_1 = z + L/2$

En resumen tenemos

$$= k \lambda_0 \times \left[\frac{v}{a^2 \sqrt{a^2 + v^2}} \right]_{z+L/2}^{z-L/2}$$

$$E_x = -k \lambda_0 \times \left[\frac{(z - L/2)}{(x^2 + y^2) \sqrt{x^2 + y^2 + (z - z')^2}} - \frac{(z + L/2)}{(x^2 + y^2) \sqrt{x^2 + y^2 + (z - z')^2}} \right]$$

La componente E_y es:

$$E_y = -k \lambda_0 y \left[\frac{(z - L/2)}{(x^2 + y^2) \sqrt{x^2 + y^2 + (z - z')^2}} - \frac{(z + L/2)}{(x^2 + y^2) \sqrt{x^2 + y^2 + (z - z')^2}} \right]$$

Ahora resolvemos lo siguiente

$$E_z = k \lambda_0 \int_{-L/2}^{L/2} \frac{(z - z')}{[x^2 + y^2 + (z - z')^2]^{3/2}} dz' = k \lambda_0 \int_{z+L/2}^{z-L/2} \frac{-v dv}{[x^2 + y^2 + v^2]^{3/2}}$$

$v = z - z'$
 $dv = -dz'$

$$v_2 = z - \frac{L}{2} \quad \wedge \quad v_1 = z + \frac{L}{2}$$

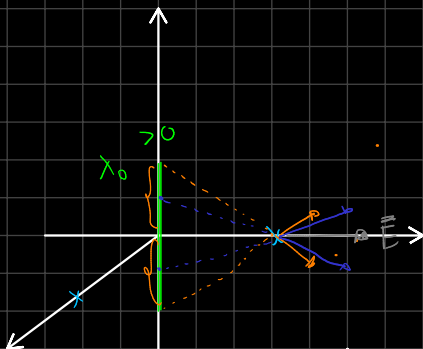
Resolvemos la siguiente integral $-k \lambda_0 \int_{z+L/2}^{z-L/2} \frac{v dv}{[x^2 + y^2 + v^2]^{3/2}}$

La primitiva será $\int \frac{u du}{[a^2 + u^2]^{3/2}} = \frac{-1}{\sqrt{u^2 + a^2}}$

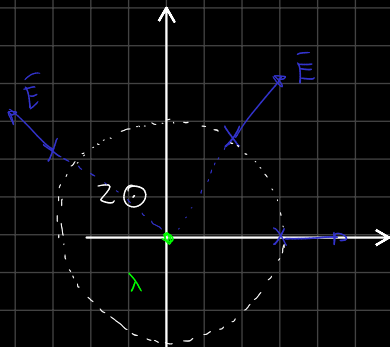
$$E_z = -k\lambda_0 \int_{z+l/2}^{z-l/2} \frac{u du}{[x^2 + y^2 + u^2]^{3/2}} = k\lambda_0 \left[\frac{1}{\sqrt{x^2 + y^2 + (z-l/2)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z+l/2)^2}} \right]$$

Tenemos lo siguiente:

$$\begin{aligned} \text{Tendremos } \vec{E}(x, y, z) = k\lambda_0 & \left[\left[\frac{(z+l/2)}{(x^2+y^2)\sqrt{x^2+y^2+(z+l/2)^2}} - \frac{(z-l/2)}{(x^2+y^2)\sqrt{x^2+y^2+(z-l/2)^2}} \right] (x\hat{x} + y\hat{y}) \right. \\ & \left. + \left[\frac{1}{(x^2+y^2+(z-l/2)^2)^{3/2}} - \frac{1}{(x^2+y^2+(z+l/2)^2)^{3/2}} \right] \hat{z} \right] \end{aligned}$$

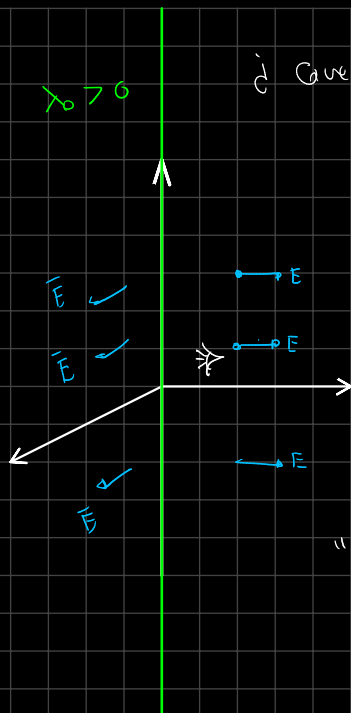


Tenemos entonces si $z=0 \rightarrow E_z=0$



Tendremos entonces

$$\vec{E}(r) = \frac{k\lambda_0}{r} \left[\frac{(z+l/2)}{\sqrt{r^2+(z+l/2)^2}} - \frac{(z-l/2)}{\sqrt{r^2+(z-l/2)^2}} \right] \hat{r} + k\lambda_0 \left[\frac{1}{(r^2+(z-l/2)^2)^{3/2}} - \frac{1}{(r^2+(z+l/2)^2)^{3/2}} \right] \hat{z}$$



¿Que pasa si lo hacemos infinito?

Seamos de antemano que

$$\underline{\text{Hilo } L} \quad \vec{E}(\vec{r}) = \frac{k\lambda_0}{r} \left\{ \frac{(z + L/2)}{[r^2 + (z + L/2)^2]^{3/2}} - \frac{(z - L/2)}{[r^2 + (z - L/2)^2]^{3/2}} \right\} \hat{r} +$$

$$k\lambda_0 \left\{ \frac{1}{\sqrt{r^2 + (z - L/2)^2}} - \frac{1}{\sqrt{r^2 + (z + L/2)^2}} \right\} \hat{k}$$

$\xrightarrow{L \rightarrow \infty}$

"Hilo" = ∞ $\begin{cases} L \rightarrow \infty \\ z, r \ll L \end{cases}$

Condición que anula la componente $\vec{E}_z$

$E_z = 0$ ($z=0$) $\forall z$ Obtenemos

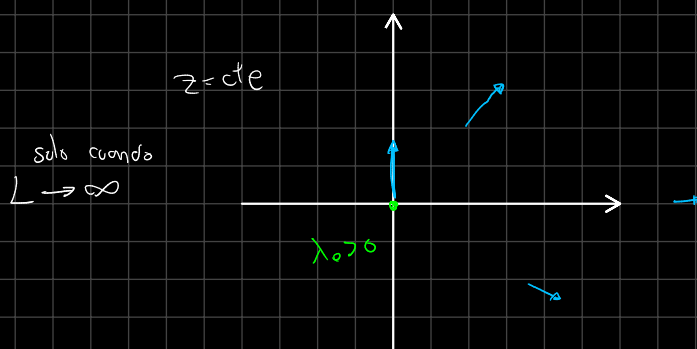
$$\vec{E}(\vec{r}) = k\lambda_0 \left[\frac{(z + L/2)}{|z + L/2| \sqrt{\left(\frac{r}{z + L/2}\right)^2 + 1}} - \frac{(z - L/2)}{|z - L/2| \sqrt{\left(\frac{r}{z - L/2}\right)^2 + 1}} \right] \hat{r}$$

$\xrightarrow{L \rightarrow \infty}$ $\xrightarrow{z, r \ll L}$

$$\vec{E}(\vec{r}) = \frac{k\lambda_0}{r} [1 - (-1)] = \frac{k\lambda_0}{r} [2] = \frac{2k\lambda_0}{r}$$

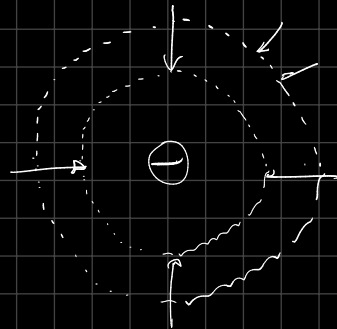
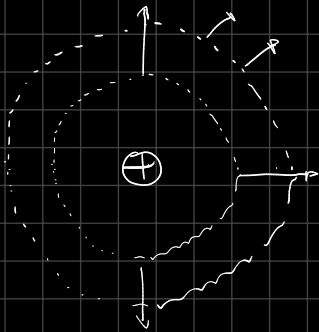
Conclusión

$$\boxed{\vec{E}(\vec{r}) = \frac{2k\lambda_0}{r} \hat{r} \quad \text{Hilo } \rightarrow \infty}$$

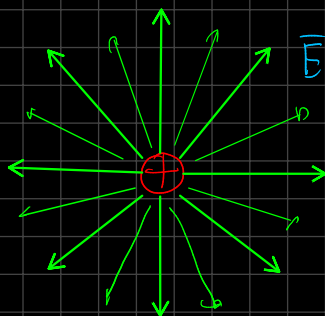
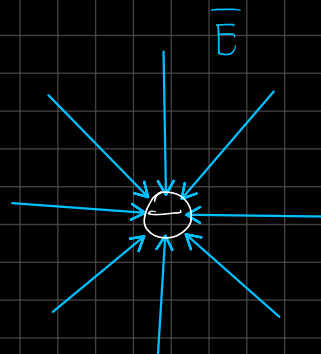
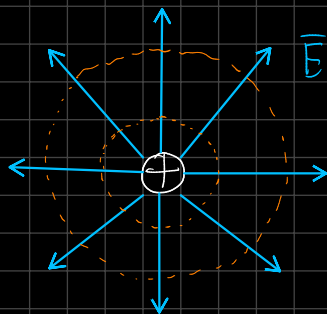
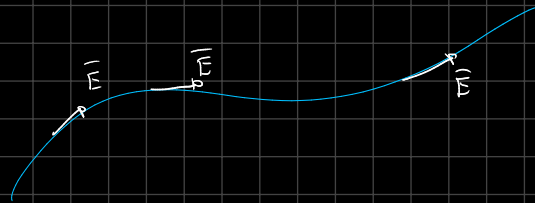


Representación de $\vec{E}$

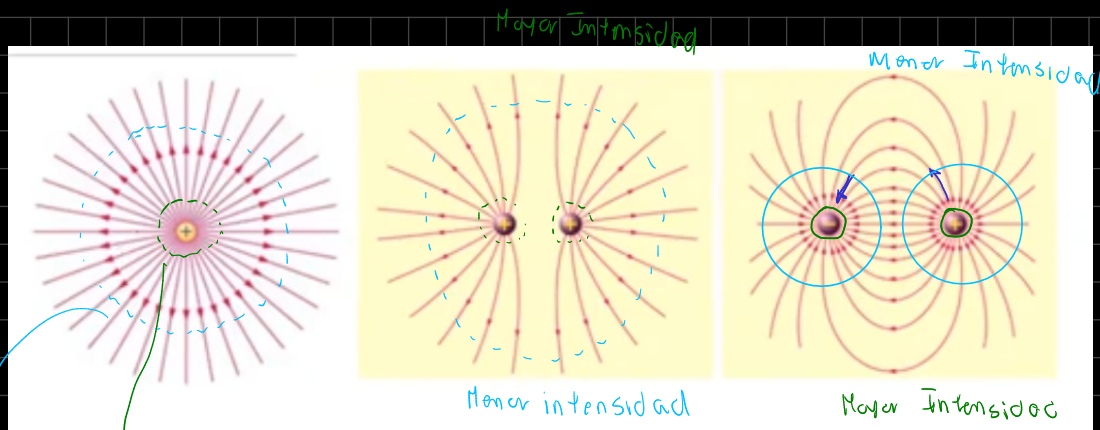
$$\vec{E}(\vec{r}) = \frac{k q}{r^2} \hat{r}$$



Lineas de Campo



¿Tanta cantidad de linea
como intensidad de esa
carga ∇_0



Mayor Densidad de líneas

Menor densidad de líneas

Cargas Iguales
ya tienen la
misma cantidad
de "líneas de campo"

$$|q_+| = |q_-|$$

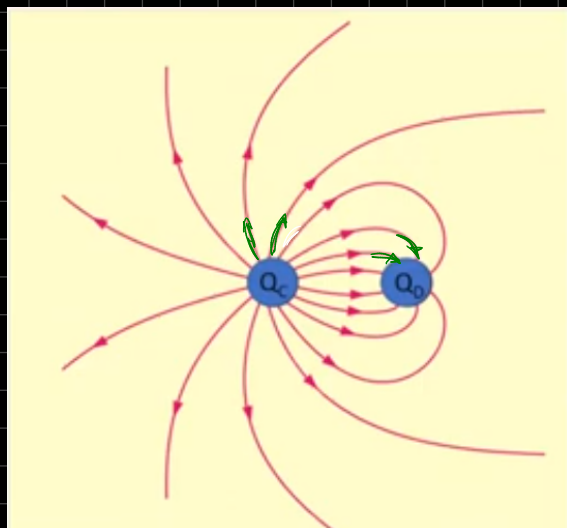
12 12

$$|q_-| = |q_+|$$

25 25
línea línea

Dipolo

Menor Densidad de líneas Menor intensidad de campo eléctrico



$$Q_c >$$

$$Q_0 < 0$$

$$|Q_c| > |Q_0|$$

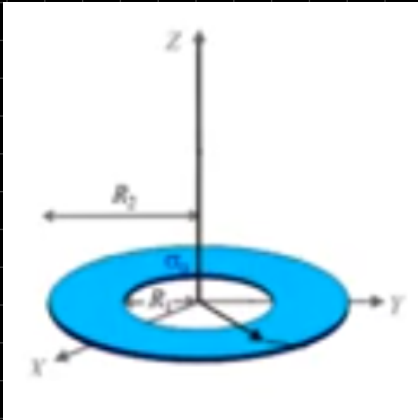
↑ ↑

16 8 Líneas
Línea

$$|Q_c| = 2 |Q_0|$$

Practica de la Clase 3

Datos: R_1 G_0 R_2



$$\vec{E}(0,0,z) = ?$$

¿ Ponemos los puntos Fuente y Campo ?

¿ Punto Campo ? $\vec{r} = (0,0,z) \Rightarrow \vec{r} = 0\hat{x} + z\hat{z}$

$\vec{r} = (x', y', 0)$ $0 < x' < R_2$
 $0 < y' < R_2$

¿ Punto Fuente ? $\vec{r}' = (r' \cos \varphi'; r' \sin \varphi'; 0)$

$\vec{r}' = r' \cos \varphi' \hat{x} + r' \sin \varphi' \hat{y} + 0 \hat{z}$

$\vec{r}' = r' \hat{r} + 0 \hat{z}$ $R_1 < r' < R_2$
 $0 < \varphi' < 2\pi$

$$\vec{E}(\vec{r}) = k \int \frac{dq' (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} \quad dq' = G_0 ds' \rightarrow ds' = r' dr' d\varphi'$$

$$\vec{E}(\vec{r}) = k \int_0^{2\pi} \int_{R_1}^{R_2} \frac{G_0 r' dr' d\varphi' (-r' \cos \varphi' \hat{x} - r' \sin \varphi' \hat{y} + z \hat{z})}{(r'^2 + z^2)^{3/2}}$$

¿ Que pasa con cada componente ?

$$E_x = k G_0 \int_0^{2\pi} \int_{R_1}^{R_2} \frac{r' dr' d\varphi' (-r' \cos \varphi')}{(r'^2 + z^2)^{3/2}} = k G_0 \int_0^{2\pi} \int_{R_1}^{R_2} (-r'^2 \cos \varphi') dr' d\varphi'$$

Tenemos que $\int_0^{2\pi} \cos \varphi' d\varphi' = \sin \varphi' \Big|_0^{2\pi} = 0$

$$E_y = k G_0 \int_0^{2\pi} \int_{R_1}^{R_2} \frac{r' dr' d\varphi' (-r' \sin \varphi')}{(r'^2 + z^2)^{3/2}} = 0$$

$\int_0^{2\pi} \sin \varphi' d\varphi' = -\cos(\varphi') \Big|_0^{2\pi} = -\cos(2\pi) - (-\cos(0)) = -1 - (-1) = 0$

Tenemos entonces la siguiente componente z

$$E_z = k G_0 \int_0^{2\pi} \int_{R_1}^{R_2} \frac{r' dr' d\varphi' z}{(r'^2 + z^2)^{3/2}} = k G_0 z \int_0^{2\pi} \int_{R_1}^{R_2} r' dr' d\varphi'$$

$$= k G_0 z \int_0^{2\pi} d\varphi' \int_{R_1}^{R_2} \frac{r' dr'}{(r'^2 + z^2)^{3/2}}$$

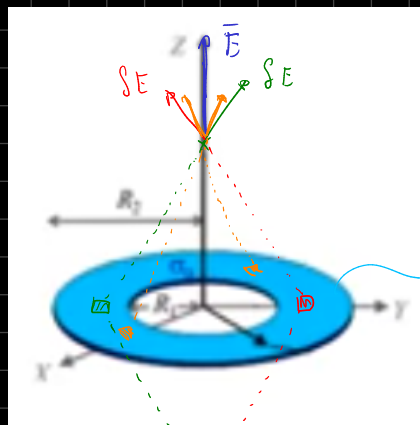
por Tabla

$$\int \frac{u du}{(u^2 + v^2)^{3/2}} = \frac{-1}{\sqrt{u^2 + v^2}}$$

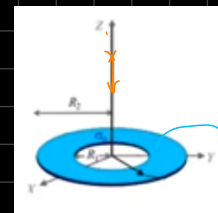
$$= k G_0 z [2\pi] \left[\frac{-1}{\sqrt{r'^2 + z^2}} \right]_{R_1}^{R_2}$$

$$= k G_0 z 2\pi \left[\frac{1}{\sqrt{R_1^2 + z^2}} - \frac{1}{\sqrt{R_2^2 + z^2}} \right]$$

$$\vec{E}(0,0,z) = k G_0 z 2\pi \left[\frac{1}{\sqrt{R_1^2 + z^2}} - \frac{1}{\sqrt{R_2^2 + z^2}} \right] \hat{z}$$



Solo sobrevive en la componente en $\hat{z}$

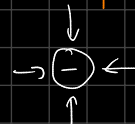


$G_0 < 0$

Tomar lo análogo de



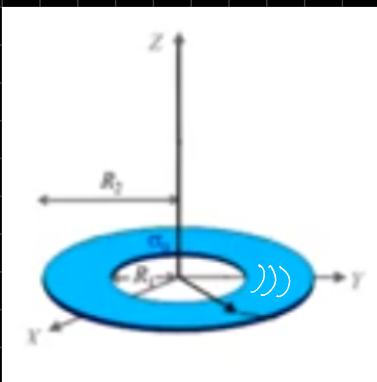
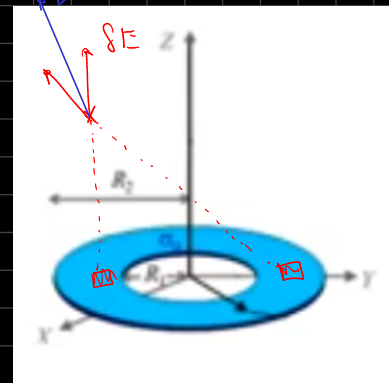
y



¿Observación? ¿Que pasa si pongo una carga en una posición donde No sea
 anula x, y ? Ejemplo $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

¿ $\vec{E}$ tendrá 3 componentes?

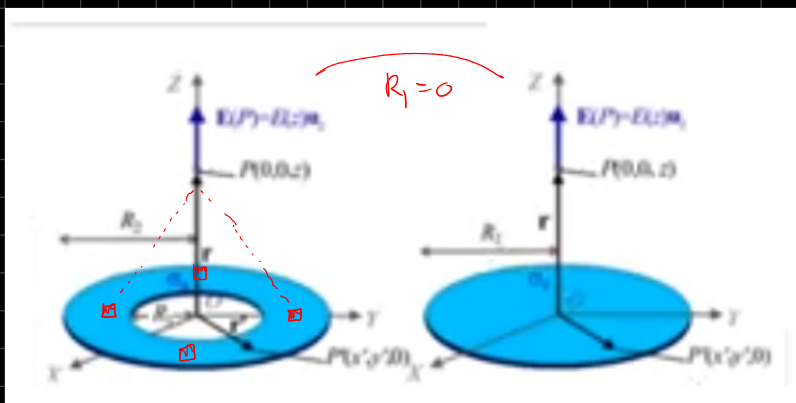
¿Y si G no es Uniforme?



$$G(r') = 2\pi' \frac{q}{m^2}$$

$$G(r') = 2\cos\phi' \frac{q}{m^2}$$

¿Y si cambiamos de Geometría?



$$\vec{r} = (0, 0, z) \quad z \neq 0$$

$$\vec{r}' = (r' \cos \phi'; r' \sin \phi'; 0)$$

$$0 < r' < R_2$$

$$0 < \phi' < 2\pi$$

$$d\phi' = G_0 r' dr' d\phi'$$

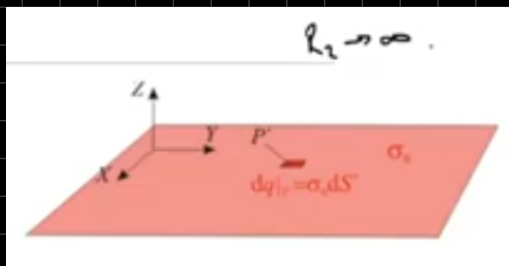
$$\vec{E}(0, 0, z) = kG_0 z 2\pi \left[\frac{1}{\sqrt{R_1^2 + z^2}} - \frac{1}{\sqrt{R_2^2 + z^2}} \right] \hat{z}$$

¿También entonces lo siguiente?

$$\vec{E}(\vec{r})_{\text{CORONA}} = k G_0 z 2\pi \left(\frac{1}{R_1^2 + z^2} - \frac{1}{\sqrt{R_2^2 + z^2}} \right) \hat{z}$$

$$\vec{E}(\vec{r})_{\text{Disco}} = k G_0 z 2\pi \left(\frac{1}{|z|} - \frac{1}{\sqrt{R_2^2 + z^2}} \right) \hat{z}$$

¿Cómo pasa si llevamos a $R_2 \rightarrow \infty$?



$$\vec{E}(\vec{r})_{\text{Disco}} = k G_0 z 2\pi \left(\frac{1}{|z|} - \frac{1}{\sqrt{R_2^2 + z^2}} \right) \hat{z}$$

$$\vec{E}_{\text{plano } \infty} = k G_0 z 2\pi \left(\frac{1}{|z|} - \frac{1}{\sqrt{R_2^2 + z^2}} \right) \hat{z}$$

$$= k G_0 z 2\pi \left(\frac{1}{|z|} - \frac{1}{R_2 \sqrt{1 + \left(\frac{z}{R_2}\right)^2}} \right) \hat{z}$$

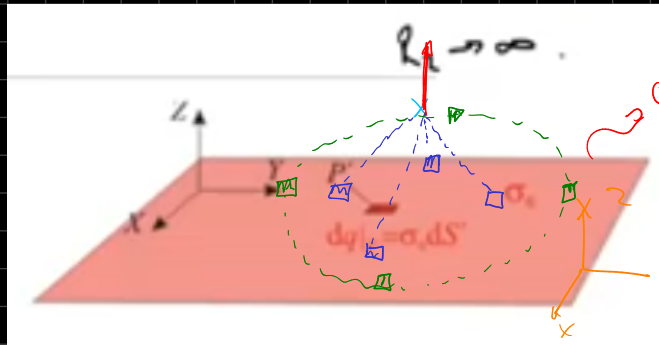
$\xrightarrow{R_2 \rightarrow \infty} \frac{1}{\infty} \rightarrow 0$
 $\xrightarrow{R_2 \rightarrow \infty} \frac{z}{R_2} \rightarrow 0$

$$\vec{E} = k G_0 z 2\pi \frac{1}{|z|} \hat{z}$$

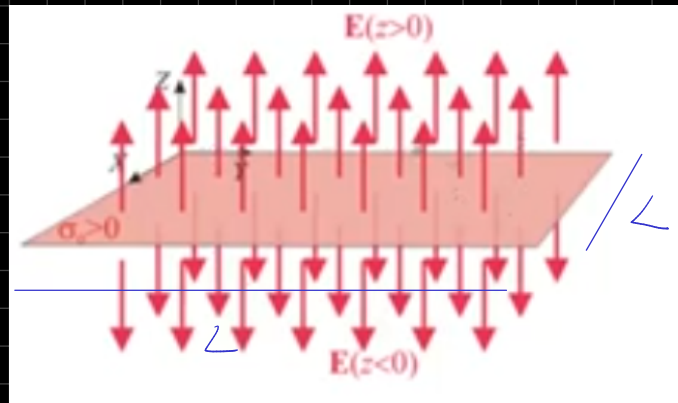
$\begin{cases} +1 & z > 0 \\ -1 & z < 0 \end{cases}$

$$\vec{E}_{\text{plano}} = \begin{cases} k G_0 2\pi \hat{z} & z > 0 \\ k G_0 2\pi (-\hat{z}) & z < 0 \end{cases}$$

$$\vec{E} = k G_0 2\eta \left(\frac{\hat{z}}{z^2}\right)$$



$$\vec{E} = k G_0 2\eta \left(-\frac{\hat{z}}{z^2}\right)$$



$$\vec{r} = (0, 0, \frac{z}{2})$$

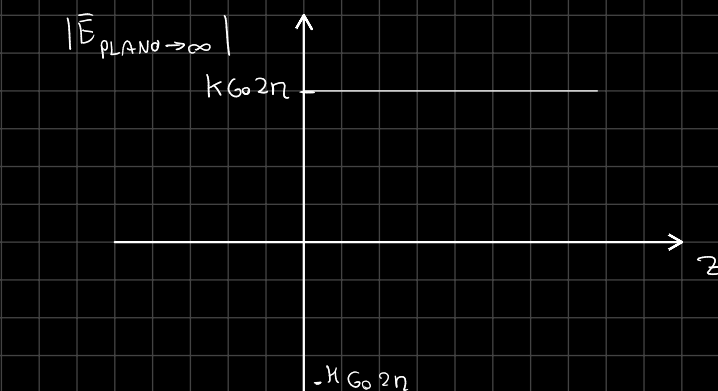
$$\vec{r}' = (x', y', 0)$$

$$-\frac{L}{2} < x' < \frac{L}{2}$$

$$-\frac{L}{2} < y' < \frac{L}{2}$$

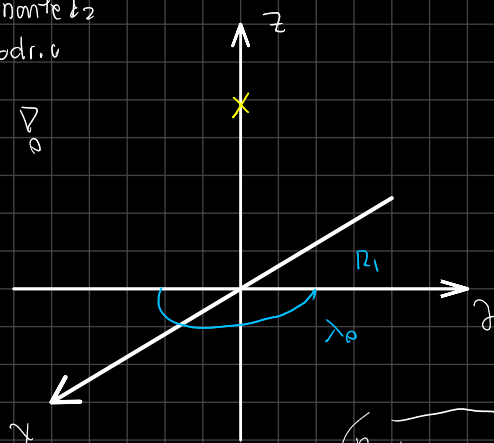
En caso de tener un plano finito hay que plantear desde
cero "Punto campo" - "Punto Fuente"

$$dq' = G_0 dx' dy'$$



Ejercicio Medido por M1

¡ Cuidado! La componente E_z no es única podr. v tener otros componentes ∇_0



$$\vec{E}(0,0,z) = ?$$

||

E_z

Punto Campo

$$\vec{r} = 0\hat{x} + 0\hat{y} + z\hat{z}$$

Punto Fuente

$$\vec{r}' = r' \cos \varphi' \hat{x} + r' \sin \varphi' \hat{y} + 0\hat{z}$$

$$\vec{r}' = R_1 \cos \varphi' \hat{x} + R_1 \sin \varphi' \hat{y} + 0\hat{z}$$

$$-\frac{\pi}{2} < \varphi' < \frac{\pi}{2}$$

Entonces

$$\vec{E}(0,0,z) = k \int dq' \frac{|\vec{r} - \vec{r}'|}{|\vec{r} - \vec{r}'|^3}$$

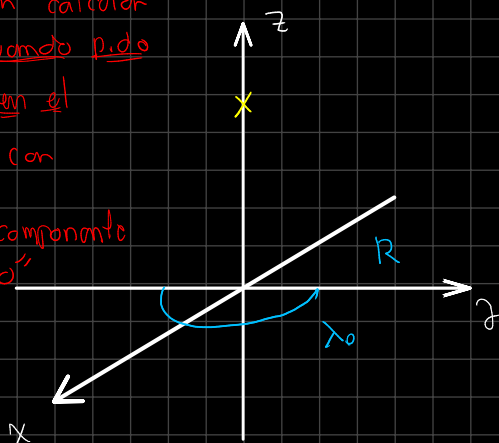
$$\vec{r} - \vec{r}' = -R_1 \cos \varphi' \hat{x} - R_1 \sin \varphi' \hat{y} + z\hat{z} = -R_1 \hat{r} + z\hat{z}$$

$$|\vec{r} - \vec{r}'| = \sqrt{R_1^2 (\cos^2 \varphi' + \sin^2 \varphi') + z^2} = \sqrt{R_1^2 + z^2}$$

$$dq' = \lambda_0 R \cdot d\varphi' \quad -\frac{\pi}{2} < \varphi' < \frac{\pi}{2}$$

¿Cuidado con calcular
"ojo con cuando pido
el campo en el
eje z" con

"Pedir la componente
z del campo"



$$\vec{E}(0,0,z) = ?$$

+

$$E_z$$

¿Como continua esto?

$$\vec{E}(0,0,z) = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$$

1º) ¿Quem es $\vec{r}$ y $\vec{r}'$?

$$\vec{r} = (0, 0, z)$$

$$\vec{r}' = (R \cos(\varphi'), R \sin(\varphi'), 0)$$

$$dq' = \lambda_0 dl' = \lambda_0 R d\varphi' \quad \text{que va en } -\frac{\pi}{2} < \varphi' < \frac{\pi}{2}$$

¿Ahora debemos armar la ley de Coulomb?

$$\vec{E}(0,0,z) = k \int \frac{dq' (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} = k \int \frac{\lambda_0 dl' (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} = k \int \frac{\lambda_0 R d\varphi' (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} = (*)$$

$$\vec{r} - \vec{r}' = (-R \cos(\varphi') \hat{x} - R \sin(\varphi') \hat{y} + z \hat{z})$$

$$|\vec{r} - \vec{r}'| = \sqrt{R^2 \cos^2(\varphi') + R^2 \sin^2(\varphi') + z^2} = \sqrt{R^2 + z^2} =$$

entonces

$$(*) = k \int \frac{\lambda_0 R d\varphi' (-R \cos(\varphi') \hat{x} - R \sin(\varphi') \hat{y} + z \hat{z})}{(R^2 + z^2)^{3/2}}$$

Entonces tendremos

$$E_x = k \int_{-\pi/2}^{\pi/2} \frac{\lambda_0 R d\varphi' (-R \cos(\varphi'))}{(R^2 + z^2)^{3/2}}$$

$$E_y = k \int_{-\pi/2}^{\pi/2} \frac{\lambda_0 R d\varphi' (-R \sin(\varphi'))}{(R^2 + z^2)^{3/2}}$$

$$E_z = k \int_{-\pi/2}^{\pi/2} \frac{\lambda_0 R d\varphi' (z)}{(R^2 + z^2)^{3/2}}$$

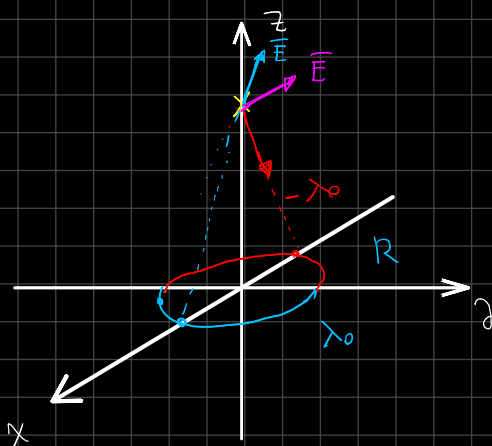
$$\int_{-\pi/2}^{\pi/2} \frac{R d\varphi' (z)}{(R^2 + z^2)^{3/2}}$$

$$\vec{E}(0,0,z) = k\lambda_0 \left[\int_{-\pi/2}^{\pi/2} \frac{R d\psi' (-R \cos(\psi'))}{(R^2 + z^2)^{3/2}} \hat{x} + \int_{-\pi/2}^{\pi/2} \frac{R d\psi' (-R \sin(\psi'))}{(R^2 + z^2)^{3/2}} \hat{y} + \int_{-\pi/2}^{\pi/2} \frac{R d\psi' (z)}{(R^2 + z^2)^{3/2}} \hat{z} \right]$$

$$\vec{E}(0,0,z) = \frac{k\lambda_0 R}{(R^2 + z^2)^{3/2}} \left[\underbrace{-R \int_{-\pi/2}^{\pi/2} \cos(\psi) d\psi}_{-R(z)} \hat{x} + \underbrace{\int_{-\pi/2}^{\pi/2} -R \sin(\psi') d\psi'}_0 \hat{y} + \underbrace{\int_{-\pi/2}^{\pi/2} z d\psi'}_{zR\pi} \hat{z} \right]$$

$$\vec{E}(0,0,z) = \frac{k\lambda_0 R}{(R^2 + z^2)^{3/2}} \left[-zR \hat{x} + zR\pi \hat{z} \right]$$

Quindi positivo se



$$\vec{E}(0,0,z) = \frac{k\lambda_0 R}{(R^2 + z^2)^{3/2}} \left[\underbrace{-R \int_{-\pi/2}^{\pi/2} \cos(\psi) d\psi}_{-2R} \hat{x} + \underbrace{\int_{-\pi/2}^{\pi/2} -R \sin(\psi') d\psi'}_0 \hat{y} + \underbrace{\int_{-\pi/2}^{\pi/2} z d\psi'}_{zR\pi} \hat{z} \right]$$

$$\vec{E}(0,0,z) \left[2R \hat{x} + zR\pi \hat{z} \right]$$